COMPUTER NETWORKS

Assignment – Model Answer Sheet

Maximum Marks: 50

Q1. **[6 Marks]**

Given

- LAN bandwidth = 10 Mbps
- Measured traffic = 12,000 frames/min, each frame = 10,000 bits

(a) Total bits per minute [2]

```
12,000 × 10,000 = 120,000,000 bits
```

(b) Throughput [2]

```
120,000,000 / 60 = 2,000,000 bps = 2 Mbps
```

(c) Why it is lower than 10 Mbps [2]

- Protocol overhead, collisions and retransmissions
- Limits in application or host processing
- Congestion or queuing delays
- Errors and retransmission reduce the effective throughput

Q2. **[7 Marks]**

(a) Byte pairs and fields [2]

```
CB 84 | 00 0D | 00 1C | 00 1C
```

The four fields are: Source Port, Destination Port, Length and Checksum.

(b) Field values [3]

- Source port = 0xCB84 = 52100 (decimal)
- Destination port = 0x000D = 13
- Length (total UDP length) = 0x001C = 28 bytes
- Checksum = 0x001C (28 in decimal)

(c) Size of UDP data [1]

UDP header = 8 bytes, so data size = 28 − 8 = 20 bytes.

(d) Byte order and Length field [1]

- Ports are 16-bit fields stored in network byte order (big-endian).
- Length is the length of the entire UDP datagram (header + data).

Q3. **[5 Marks]**

(a) Mask and network address [1]

For /24 the subnet mask is 255.255.255.0 and the network address is 17.63.110.0.

(b) Usable and broadcast addresses [3]

- First usable address: 17.63.110.1

- Last usable address: 17.63.110.254
- Broadcast address: 17.63.110.255

(c) Host range [1]

Host range is .1 through .254; .0 is the network address and .255 is the broadcast address.

Q4. **[6 Marks]**

TCP/IP layers (top to bottom)

```
Application Layer ↓ Transport Layer ↓ Network Layer ↓ Data Link Layer
```

Application Layer

- Process-to-process communication
- Application-specific semantics (e.g., SMTP mail delivery rules)

Transport Layer

- Reliable byte-stream delivery (TCP), ensuring in-order delivery
- Multiplexing/demultiplexing using ports, allowing many applications to share the network

Network Layer

- Routing: finding a path between source and destination across the network
- Logical addressing (IP addressing) and fragmentation

Data Link Layer

- Frame transmission and media access
- Error detection at frame level

Q5. **[6 Marks]**

(a) Stop-and-Wait and its inefficiency [2]

- The sender transmits one frame and then waits for the ACK.
- If the Round-Trip Time (RTT) is large compared with the frame transmission time, the channel stays idle for most of the time.

(b) Link utilization [1]

```
U = Frame transmission time / (Frame transmission time + RTT)
```

(c) Selective Repeat ARQ [3]

- The sender can transmit multiple frames at once (sliding window).
- Only the erroneous frames are retransmitted, not the whole sequence.
- This increases throughput and utilization, especially on long-delay links.

Q6. **[4 Marks]**

- CRC computes the remainder of the polynomial division of the data by a generator polynomial.
- The remainder is appended to the data as the checksum.

- The receiver divides the received bits by the same generator; a non-zero remainder means an error is detected.
- Practical result: it detects burst errors and many random errors with high probability.

Q7. **[5 Marks]**

(a) Address size [1]

IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses.

(b) Header and options [2]

IPv6 has a simplified header and uses extension headers for options, which gives faster routing.

(c) Built-in features [2]

IPv6 supports autoconfiguration and eliminates the need for NAT; this affects addressing and routing.

Q8. **[5 Marks]**

(a) DHCP server and pool [1]

- The DHCP server manages a pool of IP addresses.
- When a client requests one, the server leases an unused address.

(b) DORA [2]

`Discover` → `Offer` → `Request` → `Acknowledge`

(c) ARP probe [1]

Before assignment, DHCP may send an ARP probe to confirm the address is not already in use.

(d) Lease expiration [1]

Lease expiration is used to reclaim addresses.

Q9. **[6 Marks]**

(a) TCP vs UDP [3]

- TCP (connection-oriented): reliable and ordered, but higher latency (retransmissions, head-of-line blocking).
- UDP (connectionless): faster with minimal overhead, but no reliability.

(b) Recommendations [3]

- Real-time streaming / gaming / VoIP → UDP, because speed matters more than perfect reliability.
- On-demand video (Netflix, YouTube) → TCP/HTTP.

— End of Answer Sheet —